

# The Foundation: $(\mathbb{S}, \tau, \omega)$ and Unsharp Unity

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Opening statement for the next series. Bare minimum only.

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## 1. The triple

[DEF]  $\mathbb{S} = \mathbb{R}_\varepsilon[(\mathbb{Z}_2)^4]$ : the sixteen-dimensional real Cayley–Dickson algebra — the specific cocycle produced by iterated doubling from  $\mathbb{R}$ , not an arbitrary twisted group algebra on  $(\mathbb{Z}_2)^4$  (Clifford algebras arise from other cocycles on the same group and are not meant here). Unital, flexible, quadratic, power-associative, central simple, positive-definite norm — and, stated with the same weight, **not** associative, **not** alternative, and possessed of zero divisors: these failures are starting data too, not omissions from a list of virtues.  $\text{Aut}(\mathbb{S}) = G_2 \times S_3$ .

[DEF]  $\tau$ : an involution in the  $S_3$  factor.  $\mathbb{O} = \text{Fix}(\tau)$ ,  $\mathbb{O}_{e_8} = \ker(\tau + 1)$ . Three conjugate splits exist, related by the order-three  $\mu$ ; the choice among them is symmetric — an orbit coordinate, not a further commitment.

[FACT] The mediator direction used to build  $X, P, W$  below is, by the same logic, an orbit coordinate:  $G_2$  acts transitively on the unit sphere in  $\text{Im}(\mathbb{O})$ , so any choice of mediator is related to any other by an automorphism, and none is a further commitment.

[FACT, resting on a cited construction, not independently audited here] A dilation symmetry absent from the parent octonions is created by a specific contraction on  $\text{Im}(\mathbb{O}) = \text{Im}(\text{Fix}(\tau))$ , survives as an automorphism through the tower that follows, and its cited unique  $\hbar$ -preserving component is the squeeze flow used throughout. The mediator’s freedom is genuine and shown above; the contraction scheme itself — its grading and scaling regime — is forced *within a declared*  $1+3+3$  ansatz (one central generator, three coordinates, three momenta), exhaustively so, verified by the standalone artifact below. The ansatz itself — including, specifically, the three-dimensionality of both triples — is no longer this construction’s free input: a companion artifact (cited in the inventory below) proves it unconditionally, over every partition and grade level a contraction of  $\text{Im}(\mathbb{O})$  could take, with no dependence on any enumeration. The single hypothesis the proof needs — that the contraction retain any non-associative content at all — is not a further free choice; it is why the seed is  $\mathbb{O}$  rather than an associative algebra to begin with. This remains the single citation on which the inventory claim below, §2’s family, and  $\chi$ ’s entire domain most directly depend — that dependence is unchanged; what has closed is the citation’s free input, previously the first item on the ledger and now removed there, while the underlying papers’ locator debts remain open below.

[DEF]  $\omega$ : a regular quasi-free state, normal on its own GNS type  $\text{III}_1$  factor — a generic member of a sector family rather than one fixed point. Which member is physically realized is physical data, not part of the formal starting point. The construction of the operational algebra, and the results cited in §2, are established in a specific, named prior body of work — five papers, stated once here so later citation is unambiguous: the second-contraction paper (the  $\mathfrak{so}(3)$  reduction and the flow above); the squeeze-thermal-state paper (the type-I no-go, the scale-line construction, the type  $\text{III}_1$  verdict, and Corollary 1.2 — the  $\mathfrak{so}(3)$ –flow commutation used in §2, now proven rather than merely verified, by a one-line tensor-factor argument recorded in the standalone artifact); the centralizer paper (closing

the last construction-specific link); the uniqueness paper, *The State Is Unique* (§2, KMS uniqueness at full quasi-free strength); and the infrared write-out paper, *The Infrared Write-Out* (§6, Scope, stating  $\beta$  as the construction’s sole physical parameter — §2’s load-bearing citation). None of it is re-derived here; full locators are owed throughout, though not from nothing: section-level locators already exist for two of the five, a corollary number for a third. **A sixth document now exists alongside these:** *The Keystone: From  $(\mathbb{S}, \tau)$  to the Injective Type  $III_1$  Factor*, the standalone, sequential compilation of the full chain from these five papers into one checkable argument, previously carried on this ledger only as an aspiration. It closes one open item, on a standard argument (injectivity of the quasi-free algebra — the property Haagerup’s own theorem actually needs, its title naming *the injective factor of type  $III_1$*  — by a recorded standard argument, stopping short of the stronger and unnecessary “hyperfinite,” still awaiting a second reader’s confirmation per the ledger below) and, together with a seventh document it cites as a companion (*The Non-Associativity Selector*, proving the 1+3+3 grading shape unconditionally — see §1’s fourth [FACT] above), closes what it first only located: the ansatz itself is no longer this chain’s free input. The second-contraction paper’s own locator remains the most urgent single citation debt among the five, since it is the least pinned down of the underlying set. The checking done on these results within this collaborative process is real but is not independent peer review or formal proof-checking.

[SCOPE P-Em] The order of deduction is not the order of being.  $\mathbb{S}$  is the persistence-form of a ground outside all algebraic and metric vocabulary;  $\omega$  is that ground’s sole imprint — one kind of object mediates the ground’s presence, not one formally-singled-out instance of it; which member of the family is realized is empirical input, not formal data. Every claim below is scoped *among representable structures*. This does no deductive work — nothing is derived from it. Its function is to scope every other claim in this document.

$(\mathbb{S}, \tau, \omega)$  is the whole of the formal beginning. No further formal data is added; the realized sector is empirical input.

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## 2. Unsharp unity

Two claims, of different logical kinds, both necessary, neither to be confused with the other.

[FACT] The  $\mathfrak{so}(3)$  remnant arises in two steps on  $\mathbb{O} = \text{Fix}(\tau)$ , the octonion algebra where compositionality genuinely holds, acting through its seven-dimensional imaginary part  $\text{Im}(\mathbb{O})$ : fixing the mediator direction alone gives  $\mathfrak{su}(3)$ , the classical stabilizer of a point on  $S^6$  ( $G_2/SU(3) = S^6$ ); additionally fixing the  $P$ -triple as a real subspace compatible with the resulting complex structure reduces this to  $\mathfrak{so}(3)$ . The reduction and its dimensions rest on a symbolic argument — derivations of a composition algebra are skew with respect to the norm form — with machine-precision checks in the same paper confirming specific instances on top of it, not in place of it. This same  $\mathfrak{so}(3)$  survives as an exact automorphism through the entire contraction tower. That it preserves  $\omega$  itself, rather than merely mapping invariant states to invariant states, follows without a separate regularity-stability lemma:  $\mathfrak{so}(3)$  acts by Bogoliubov automorphisms, which preserve quasi-freeness by definition and are proven to commute with the derived flow exactly — Corollary 1.2 of the squeeze-thermal-state paper, with machine-precision numerics confirming the proof, not substituting for it;  $\omega$  composed with any such rotation is accordingly a quasi-free  $\beta$ -KMS state of the same flow at the same temperature, and the uniqueness theorem — proven at full quasi-free strength (not merely among a restricted “regular”

subclass), *The State Is Unique*, §2 — forces it to equal  $\omega$ .

**[PROP, proven within any established folium; open across the family]** No admissible configuration is flowless: by the freeze dichotomy, flow trivial  $\iff$  tracial, and no normal tracial state exists — this holds for every state normal to a folium of established type, in particular  $\omega$ ’s own. No trace exists at all, not even a normal semifinite one, on the folium’s von Neumann algebra; its modular spectrum locks the type specifically to  $\text{III}_1$ . (Non-faithful normal states are covered via their support corners, which remain type  $\text{III}$ .) **Open:** whether this type verdict holds uniformly across every member  $\beta$  of the family below, or has been established only at  $\omega$ ’s own parameters — parameter-uniformity is its own ledger entry, not folded into this proposition.

**[DEF]** The symmetric reference is the pure vacuum of the unsqueezed, zero-temperature configuration — the corpus’s own earlier language for the unity horizon, “the free scaffold vacuum.” It is a factor state, as is any genuinely thermal member of the family; for factor states, quasi-equivalence and disjointness are the only two possibilities. Its disjointness from any thermal member needs no covariance estimate: quasi-equivalent representations generate isomorphic von Neumann algebras, and isomorphism preserves type; the reference, being pure, generates a type I folium ( $B(H)$ , acting irreducibly). Disjointness from  $\omega$ ’s own folium is accordingly unconditional, since that folium’s type is established directly. Disjointness from every other family member holds too, but currently inherits the same conditionality as the [PROP] above — pending only that the member’s folium is *not type I*, which is strictly weaker than pinning  $\text{III}_1$  specifically: any non-type-I folium already forbids isomorphism with  $B(H)$ , hence forbids quasi-equivalence, hence gives disjointness by the dichotomy, without needing the full Connes invariant. This decomposes the parameter-uniformity ledger item into a cheap half (not type I, sufficient here) and an expensive half (the full  $\text{III}_1$  verdict, needed elsewhere for the modular facts) rather than charging this exclusion for the whole thing.

**[PROP, proven, the endpoints distinguished]** The operational algebra is CCR/Weyl, not CAR. Bosonic quasi-free covariance satisfies  $\sigma \geq \frac{1}{2} \cdot \text{Id}$ , with equality exactly at the pure, unsqueezed vacuum —  $\frac{1}{2}$  marks the *symmetric* end, not the settled one. The trace is not a quasi-free state at all: it is the non-regular counting trace on the Weyl algebra, with no covariance operator and no continuous field operators in its GNS representation. The two endpoints are distinct for a categorical reason, not a spectral one — one is a regular quasi-free state, the other is not in that class to begin with.

**[FACT]** States differing in global inverse temperature are unconditionally disjoint: extremal KMS states of the same dynamics at distinct temperatures are mutually disjoint — a standard structural theorem (Bratteli–Robinson, *Operator Algebras and Quantum Statistical Mechanics II*, §5.3, “KMS-States”; the specific theorem number within that section is not pinned here), not a computation, and independent of any covariance estimate. The theorem’s clean form presumes the states themselves are not tracial: for nontrivial dynamics, no state is KMS at two distinct temperatures, and tracial states are exactly the edge case that sits at the trivial-dynamics degeneracy, KMS at every  $\beta$  at once. That proviso is discharged automatically by the [PROP] above, which already excludes normal tracial states from these folia; the no-trace fact is what licenses the very disjointness theorem  $\chi$ ’s well-definedness leans on. This is the complete picture: the family carries no squeeze parameter independent of  $\beta$  (below), so  $\beta$ -difference is the only way two family members can differ at all, and this theorem covers that case unconditionally. The squeeze-thermal family is accordingly not a set of normal states of one von Neumann algebra; each member generates its own type  $\text{III}_1$  folium. (The name describes what the family is thermal *for* — equilibrium states of the squeeze flow, at varying temperature — not a two-parameter space of squeeze-and-temperature; one parameter,  $\beta$ , named for the dynamics it

equilibrates.)

[DEF] Admissible configuration means any state normal to *some* family member’s folium — the union of folia, with the quasi-free family serving as a transversal that parametrizes them, not as the whole of what is admissible.

[DEF,  $\chi$  as sector label]  $\chi$  extends from the family to the full union by constancy on each folium: distinct values mark disjoint GNS representations, not distinct expectation values within one. This extension is now unconditionally well-defined: with squeeze confirmed not to be an independent parameter of the family (above), the only case that could threaten well-definedness — two family members sharing a folium — reduces to distinct  $\beta$ , and the KMS-disjointness theorem above secures that unconditionally within a nontrivial-dynamics, non-tracial family, which this one is. Connes–Størmer homogeneity holds within a folium; it is precisely why a genuine order parameter has to distinguish between them, not a threat to  $\chi$ ’s existence.

[DEF,  $\chi$ ]  $\chi$  is read from the *symplectic spectrum* of the covariance operator, not the covariance itself. Symplectic eigenvalues are invariant under every symplectic transformation, squeezing included: every point of the pure stratum — zero temperature — collapses to the single value  $\nu = \frac{1}{2}$ , and thermal covariance gives  $\nu(\beta) = \frac{1}{2} \coth(\beta k/2)$  per mode ( $k$  the mode frequency, kept distinct from  $\omega$  the state throughout this document), strictly between  $\frac{1}{2}$  and  $\infty$ . (This squeeze-independence is now a fact about the invariant’s mathematics rather than a live feature of the family: the family itself carries no squeeze parameter to be independent of — see above.) The pure stratum is one boundary value; the non-regular tracial stratum is the  $\beta \rightarrow 0$  limit; the family’s closure adds both without ever containing either. [TARGET] The one piece not yet settled is uniformity in the thermodynamic limit: if the scale-line spectrum reaches  $k_n \rightarrow 0$  for the  $n$ -th mode, the per-mode invariant diverges there at every temperature, and the correct global invariant is per-mode or spectral-density-weighted rather than a bare supremum. This is the target’s single stake.

[FACT] Because  $\chi$  is built from symplectic eigenvalues, it is exactly conserved by any symplectically implemented dynamics whatsoever — the derived squeeze flow included, for every state it transports, not only at equilibrium. This is the correct signature of a legitimate sector label, not a defect of the definition. The derived flow is, at most, the dynamics internal to a fixed  $\chi$ -value; it is not, and structurally cannot be, the mechanism that changes  $\chi$ . A second, independent fact, needing nothing from the conservation theorem: every family member is individually stationary under the derived flow, being its unique KMS state at its own temperature (below), so the flow moves nothing between any two family members,  $\chi$ -equal or not — equilibrium under that flow is what a family member *is*. Together: settling — any actual change in  $\chi$ , or any motion between family members at all — is necessarily *non-Bogoliubov*, dissipative or open-system or restriction-based, and no such mechanism is currently in this document’s inventory.

[FACT, cited precisely] The squeeze-thermal family has no parameter independent of  $\beta$ . Stated directly in the cited corpus: the infrared write-out paper (*The Infrared Write-Out: Test Space, Weyl Algebra, and the Verified KMS Structure on the Scale Line*, §6, Scope) states that  $\beta$  “remains the one genuine physical parameter of the state”; the uniqueness paper (*The State Is Unique*, §2, “Uniqueness within the full quasi-free class”) proves this at full strength — the KMS condition forces any anomalous pair-correlation content (the technical form squeezing takes) to zero within the *entire* quasi-free class, not a restricted subclass. No second parameter exists; there is nothing for  $\chi$  to vary across at fixed temperature.

**[OPEN]**  $\text{Aut}(\mathbb{S}) = G_2 \times S_3$ ; the above concerns only the  $G_2$ -descended remnant.  $\mu$  acts on the same Weyl  $C^*$ -algebra — not merely an unverified assumption but, at the test-space level, literally false: the bosonic construction quantizes  $\text{Im}(\text{Fix}(\tau))$  specifically, a seven-dimensional subspace of the fifteen-dimensional  $\text{Im}(\mathbb{S})$ , and the three  $\tau$ -choices quantize three *different* such subspaces, overlapping only in whatever the full  $S_3$  group fixes jointly. The three constructions can at best be abstractly isomorphic, not literally identical. This upgrades the question into two small, finite-dimensional computations, neither done here: the dimension of the  $S_3$ -invariant intersection, which fixes how much the three constructions genuinely share; and, on that shared core, whether  $\mu$  commutes with the flow or intertwines three copies of it. If  $\mu$  is symplectically implemented on the shared core and commutes with the derived flow there, the same full-strength uniqueness argument used for  $\mathfrak{so}(3)$  gives more than  $\chi \circ \mu = \chi$ : it gives  $\mu\omega_\beta = \omega_\beta$  outright, fixing every family member, not merely preserving  $\chi$ . But there is a structural reason to expect the opposite: the flow is built from  $\text{Im}(\text{Fix}(\tau))$  for one specific  $\tau$ , while  $\mu$ 's own role is to cycle between the three conjugate choices — so  $\mu$  should *intertwine* three distinct,  $\tau$ -anchored flows rather than commute with any single one. If so, the natural picture is three separate families, one per  $\tau$ , with  $\mu$  mapping between them rather than acting within one — and settling's selection of a specific  $\tau$  becomes the expected, physical outcome, not a failed symmetry check: the same “non-uniformity would be structure” pattern already on this ledger. Commutation forces full fixing; intertwining forces physical split-selection; neither branch is decided by this sharpening, which only fixes what the question is actually asking. More generally, this is one instance of a question covering every orbit coordinate §1 declares free — the split  $\tau$  and the mediator direction alike: does the settled configuration select a specific representative, physically, of a symmetry the formalism treats as free?

**[POST, motivated, not proven]** The near-symmetric end is placed, specifically, toward the direction later identified as the past — not the future, not both alike; a zero-parameter boundary condition, with precedent in the Weyl-curvature-hypothesis tradition, naming a choice between two provably distinct ends. Global sectors are disjoint, but disjoint sectors are typically *locally* quasi-equivalent — local normality is the standard situation — so restriction to local subalgebras of a net is not merely a resource beyond the bare triple; it is the canonical place cross-sector comparison, and with it any monotonicity of  $\chi$ , becomes possible at all. Given the conservation fact above, it is also the canonical place *any* mechanism of settling would have to live, since the derived flow structurally cannot supply one.

**[SCOPE]** This section is bosonic throughout. A successful fermionic construction (§3) adds a CAR sector, and fermionic endpoint geometry runs opposite to bosonic — the CAR trace is a regular quasi-free state interior to the covariance range, with pure states at the extremes. On a combined algebra, “the settled end” is not the single thing described here; that reopening is a declared dependency of §3 succeeding, not a present defect.

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### 3. The seed, stated as a commitment

**[FACT]** No nonzero real element of  $\mathbb{S}$  can supply a nilpotent structure by serving *directly* as the Clifford/CAR product — shown independently by the algebra's positive-definite norm and by a general trace identity on its left-multiplication operators. This is not the obstacle a fermionic construction actually faces: the standard route (self-dual CAR, the same family Majorana fermions are built from) needs only Clifford relations over a real quadratic space, which  $\mathbb{S}$ 's positive-definite norm supplies un-

conditionally, once the Clifford product is built abstractly from the quadratic form rather than borrowed from  $\mathbb{S}$ 's own multiplication.

**[REFRAMED]** The open question is where the *polarization* comes from — the choice of complex structure turning the abstract, unconditionally-existing real Clifford algebra into a physical Fock representation with genuine, nilpotent, particle-like creation operators. Nilpotency is a feature of the chosen representation, not of the underlying algebra. Live candidates: a state-selected dynamics (this document's working choice); a seed complexified from the outset; the zero-divisor structure of  $\mathbb{S}$  itself; or declining a particle interpretation and working with the self-dual algebra directly.

**[COMMITMENT]**  $\mathbb{S}$  is kept real; the polarization is expected to come from  $\omega$  rather than be adjoined from outside. This is motivated by a mechanism already working for the bosonic sector: a real classical system, a derived flow, a Kay–Wald one-particle structure yielding a genuine complex Hilbert space, no adjunction. Once a state exists, Tomita–Takesaki theory supplies its own modular flow, guaranteed for any faithful normal state and certifying nothing further by itself; whether that flow agrees with the dynamics used to build the one-particle structure is the KMS property, verified per construction — what the bosonic sector's own proof supplies, not a general consequence of Tomita–Takesaki. The state-selected route is the working choice on precedent, not on necessity.

**[FACT]** If the operational algebra lands, as intended, on the hyperfinite type  $\text{III}_1$  factor, Haagerup's uniqueness theorem means its bare isomorphism class carries no memory of  $\mathbb{S}$ ,  $\tau$ , or the specific cocycle — among *hyperfinite* such factors, every one is the same algebra up to isomorphism (non-injective type  $\text{III}_1$  factors, such as the free Araki–Woods factors, are different objects, not covered here). Everything distinctive lives in the decorations: the state  $\omega$  and the sector structure it belongs to, the  $\mathfrak{so}(3)$  remnant's action, and — prospectively, contingent on §3's fermionic construction succeeding — the inclusion the  $\text{Fix}(\tau)$  half would generate within the combined algebra that construction would produce. In the current, purely bosonic operational algebra, built entirely from  $\text{Im}(\text{Fix}(\tau))$ , no proper  $\tau$ -inclusion exists at all: the subalgebra “generated by  $\text{Fix}(\tau)$ ” coincides with the whole algebra, since nothing else was ever quantized — a trivial inclusion, carrying no classification information. The genuine inclusion, if it exists, is a fermionic-sector object not yet built, not a bosonic-sector fact already in hand. This raises the stakes on §3's fermionic program beyond its existing role as the seed commitment's own condition of revision: it is now also the sole route to this document's most information-bearing prospective decoration, with nothing to replace it if that program does not succeed.

**[FACT]** Where  $\mathbb{S}$ 's actual multiplication does irreplaceable work — not its automorphism group, not its norm alone, since every 16-dimensional positive-definite form is isomorphic and carries no sedenion-specific content by itself:  $J = L_{e_4}$ , left multiplication by the mediator direction, serves directly as the complex structure in the  $\mathfrak{so}(3)$ -remnant derivation. This is one confirmed instance, not a complete inventory.

**[CONDITION OF REVISION]** This commitment holds provided the polarization can be supplied by a state-selected dynamics — a real Clifford structure over  $\mathbb{S}$ 's own quadratic form, a state-selected one-particle structure, matched against the published algebraic construction of fermion content. That construction is not complete. If it is shown that no such polarization reproduces the required content, the live alternatives above — a complexified seed foremost — become the honest fallback.

**[FACT, honestly carried]** Every falsifiability checkpoint this document or its companions name — the  $\text{III}_1/\text{III}_\lambda$  fork (REG-8), the mass-sector obligations (REG-9), the settling mechanism's own predictions (REG-2), the fermionic program's — sits behind a sector not yet constructed. The roadmap to each

is explicit; today, the framework is functionally unfalsifiable, a status the mass-sector obligation, the settling mechanism, and the fermionic program are each independently required to change, and none of the three is optional scenery: without at least one of them landing, this remains a consistent formal structure rather than a tested physical claim.

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#### 4. Ledger

Registry items this document depends on: REG-1 (injectivity, second-reader confirmation), REG-2 (mechanism of settling, non-Bogoliubov, not currently supplied), REG-4 (the  $\mu/\chi$  question), REG-5 (locator debts for companion papers and cited theorems), REG-7 (thermodynamic-limit uniformity of  $\chi$ 's symplectic invariant, §2, [TARGET]), REG-8 (the  $\text{III}_1/\text{III}_\chi$  fork), REG-9 (the mass-sector consistency check). Local to this document, roughly in order of priority: parameter-uniformity of the type  $\text{III}_1$  verdict across the family (§2) — decomposed into a cheap half (not type I, sufficient for the reference's exclusion above) and an expensive half (the full  $\text{III}_1$  verdict, needed for the modular facts elsewhere) — non-uniformity would not be a failed check, since it would mean richness is itself a function of settling depth, a structural discovery rather than a defect; the direction postulate (§2, [POST]); the fermionic condition of revision (§3); the reopening of §2's endpoint geometry if §3 succeeds (§2, [SCOPE]); a three-part verification, contingent on the fermionic construction succeeding (§3, [FACT] above): (i) that a proper  $\tau$ -inclusion is actually generated once the fermionic sector exists, rather than trivially coinciding with the ambient algebra as it does now; (ii) factoriality of that subalgebra, a question about the generating subspace's position relative to the representation's modular structure, not about state entanglement; (iii) existence of a normal conditional expectation onto it, the additional prerequisite Kosaki's index requires for type III inclusions; a complete accounting of where  $\mathbb{S}$ 's own multiplication does irreplaceable work beyond  $J = L_{e_4}$ .

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*Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.*